



REVIEW ARTICLE

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Insights into the regeneration of skin from *Acomys*, the spiny mouse

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Abstract

Members of the *Acomys* genus, known as spiny mice, are unique among mammals in being perfectly capable of regenerating large areas of skin that have been removed. During this regenerative process hairs, sebaceous glands, erector pili muscles, adipocytes and the panniculus carnosus all regenerate and the dermis does not scar. We review here the processes that the epidermis and the individual components of the dermis undergo during spiny mouse regeneration as well as the molecules that have been identified as potentially important in regeneration. We then relate this to what has been proposed as playing a role in studies from the laboratory mouse, *Mus musculus*. Differences in the immune systems of spiny mice and laboratory mice are also highlighted as this is suggested to play a part not only in the perfect wound healing that embryos display but also in regeneration in lower vertebrates.

KEYWORDS

Acomys, hair regeneration, panniculus carnosus, scarring, skin regeneration, spiny mouse

1 | INTRODUCTION

Adult mammals do not regenerate their skin and instead only heal full-thickness wounds with a hairless, smooth epithelium overlying a uniform collagenous dermis consisting of highly cross-linked parallel collagen bundles. This is the prevailing view of the result of a variety of types of damage—physical removal, surgery, tearing, chemical or thermal burning—as evidenced by our own bodies which accumulate such wounds throughout life. That is also the case for the typical laboratory mammal—rats and mice^[1]—which also heal skin damage with a hairless scar. However, this may not necessarily be the case for all mammals and it is perfectly possible that skin regenerative ability has sporadically evolved, perhaps as an adaptive response to predation rather like the regenerative ability of the lizard tail. Alternatively, regenerative ability may be a fundamental property of all animals and has undergone a widespread loss in the animal kingdom because it is neutral or detrimental to survival, but it may have been sporadically retained in the occasional adult mammal.

In support of the latter argument, the embryos of all mammals that have been investigated (sheep, monkey, mouse, rat, rabbit, human) can regenerate their skin scar-free up to the third trimester^[2] reminding us that regenerative ability may indeed be a property of all animals but it has been lost during that terminal developmental period. Some mammals, however, may not have lost this embryonic regenerative ability of their skin and carry it on into adulthood since there are some reports in the literature of good skin regeneration. The skin or velvet which grows anew each year over the new antlers of deer is perfectly formed consisting of epidermis, dermis, glands and hairs (although apparently without erector pili muscles).^[3,4] The skin of hibernating American black bears regenerates scar-free and apparently with some new hairs.^[5] A hole made through the ears of rabbits can regenerate including the overlying skin^[6,7] and the same is true for chinchillas, cows and pigs.^[8] If regenerative ability was a property of all embryos and it has been mostly lost then it may well be possible to induce it again in adults which provides positive impetus for the search for inducing molecules which could induce skin regeneration in humans. Indeed, there are good examples of identified factors which can induce hair follicles in laboratory mice because

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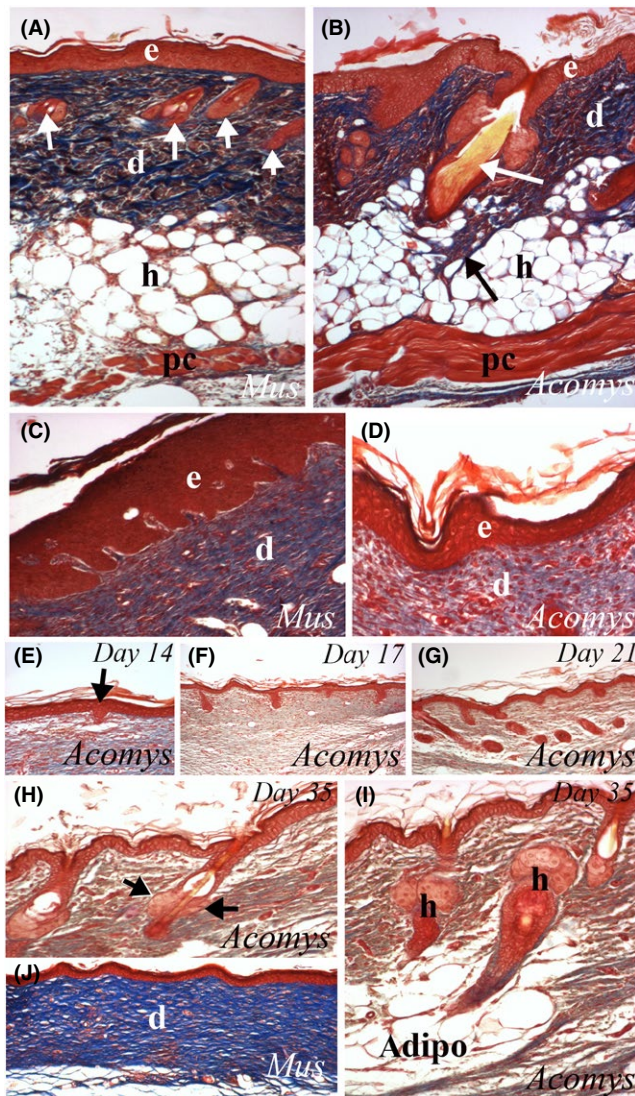


FIGURE 1 Trichrome stained sections of normal and regenerating skin. A, Normal *Mus musculus* skin showing the different layers consisting of epidermis (e), dermis (d), adipocytes in the dermal adipose tissue (h) and panniculus carnosus (pc). The relatively short hair follicles (white arrows) penetrate only about half way into the dermis. B, Normal *Acomys cahirinus* skin showing the same layers as A, but in slightly different proportions. The dermis (d) is generally thinner, the pc is thicker and the large spiny hairs penetrate through the dermis and into the hypodermis (white arrow). There are tongues of connective tissue (black arrow) which penetrate into the hypodermis perhaps to support the large spiny hairs. C, Day 21 *M. musculus* epidermis over the wound revealing its thickness and the indentations on the inner surface. The dermis (d) is stained deep blue suggesting a high level of collagen in the matrix. D, Day 21 *A. cahirinus* wound epidermis showing a thinner epithelium (e) and a less intensely stained dermal matrix (d). E–I, Stages of hair neogenesis in the *A. cahirinus* wound epithelium. E, Day 14 the first appearance of new hair follicles at the placode stage (arrow). F, Day 17 at the hair germ stage. Note the pale staining wound dermal tissue. G, Day 21 at the hair peg stage. H, Day 35 a mature hair follicle in the regenerated wound with sebaceous glands (arrows). I, Day 35 mature hair follicles (h) in the wound with adipocytes (adipo) differentiating at the base. J, Day 24 *M. musculus* wound showing a wound epithelium with no hairs present and a deep blue dermis (d) suggesting a highly collagenous matrix is forming the scar

2 | THE NORMAL SKIN

The normal skin of *Acomys* is, as a whole structure, approximately the same thickness as the skin of *Mus musculus*, but the individual layers vary somewhat. In Figure 1 is shown a comparison of the dorsal back skin of these two species. The epidermis of *Acomys* is slightly thicker than *M. musculus* because it has more cell layers whereas the dermis of *M. musculus* is thicker than that of *Acomys*. There is a thicker adipose layer in *Acomys* in the dermal adipose tissue and the skeletal muscle layer of the panniculus carnosus is more prominent in *Acomys*. The large spiny hairs of *Acomys* are the most obvious difference between these two organisms and they penetrate deeply into the hypodermis. Perhaps to support these large spiny hairs there are connective tissue projections from the dermis through the hypodermis to the panniculus carnosus. But what is most different between the two is not obvious in any physical comparison, instead it is that the skin of *Acomys* is very weak. It has a tensile strength that is only 1/20th of that of *M. musculus* skin of the Swiss Webster strain.^[17] This weakness is very obvious when *Acomys* are housed together—they occasionally fight and the skin easily tears or large areas of missing skin are generated. Presumably this is an evolved response to predation—the inherent weakness of the skin means that large pieces of skin can be torn off which allows the spiny mouse to escape. We know nothing about the mechanistic basis of this low tensile strength, but there are certainly differences in the collagen content of *Acomys* vs *M. musculus* dermis (*Mus* higher COLI and COLIV, *Acomys* higher COLIII, C-M. Chung, unpublished data). Our current proteomic analyses will hopefully provide more information in this regard. Of course, this

it transpires that large wounds made in young C57BL/6 mice do generate hair follicles although this is very variable in different strains of mice.^[9] This phenomenon, called wound induced hair follicle neogenesis (WIHN)^[10] is inhibited by prostaglandin D2 signalling^[9] and disruption of Wnt secretion.^[11] It requires *Msx2* expression,^[12] TLR3 expression^[13] and TNF expressing macrophages.^[14] It can be stimulated (produce more hairs) by a variety of methods—stimulating the Wnt pathway via disrupting the CXXC5-dishevelled interaction,^[15] over-expressing *Wnt7a*^[10] or setting up a positive feedback between FGF9 and Wnt.^[16] It is interesting to note that WIHN is not observed in the rat, a closely related rodent species.^[1]

We have recently identified an adult mammal which can perfectly regenerate its skin including the hairs in a scar-free manner after a variety of damage.^[17–20] This occurs in several species of the genus *Acomys*, the spiny mouse, and below we describe the response of the individual tissues in the skin to damage, some molecules which may potentially be involved and how these relate to WIHN in the laboratory mouse.

would not be a successful strategy if the skin did not regenerate and this is precisely what happens.

3 | SKIN REGENERATION AFTER FULL-THICKNESS EXCISION OR BURN INJURY

As a non-regenerating, scarring comparison for the regenerative ability of *Acomys cahirinus* skin we use the most outbred strain (to resemble a wild-type *M. musculus*) of laboratory mouse readily available, the CD-1. Wounds made in this strain do not undergo WIHN which varies considerably among other inbred strains.^[9] We utilize full-thickness excisional or thermal wounds and employ a range of techniques, including RT-qPCR, ELISA, immunocytochemistry, Western blotting and RNA-Seq. Our RNA-Seq analyses include a novel computational approach to directly compare expression between species by focusing on highly homologous high-scoring segment pairs along transcripts from *M. musculus* and *A. cahirinus*.

3.1 | Response of the epidermis

The rate of re-epithelialization is difficult to measure in vivo because of the variable amount of blood loss and scab formation in the wounds of both species, but in terms of the time of scab loss and measurements from histological sections *A. cahirinus* seems to be faster. In order to overcome this problem, we have measured the rate of epithelial migration in cultured keratinocytes for the two species following scratching. This experiment clearly shows that the rate of epithelial migration of *A. cahirinus* is 2× faster—*A. cahirinus* keratinocytes migrated at 20,504 $\mu\text{m}^2/\text{h}$ and CD-1 at 9011 $\mu\text{m}^2/\text{h}$.^[21]

Following skin removal there is a strong (5–8-fold stimulation) proliferative response in the keratinocytes of both species in a halo around the wound margin which causes the epithelium there to thicken, but the migrating keratinocytes themselves initially show almost no proliferative response suggesting migration and proliferation are incompatible. As wound healing progresses the proliferation rate in the migrated epidermis increases to levels seen in the surrounding wound margin.

In the third week, after wounding differences in the CD-1 and *A. cahirinus* epithelium become clearly apparent (Figure 1C,D). The CD-1 wound epithelium has become considerably thicker and indented on its lower surface (Figure 1C). The proliferation rate of the CD-1 wound epithelium now drops precipitously. In contrast, the *A. cahirinus* wound epithelium is thinner (Figure 1D) and the proliferation rate remains high because it is generating new hair follicles. Hair follicle neogenesis begins at the end of the second week with the first appearance of placodes (Figure 1E) which are undergoing Wnt signalling because the transcription factor LEF1 can be detected in their nuclei.^[17] One major gene expression difference we have identified between CD-1 and *A. cahirinus* wounds is the Wnt pathway and the expression levels of *Wnt7a* rise 40-fold in *A. cahirinus* on day 7 whereas they fall in CD-1. Although we have not shown that this gene induction occurs specifically in the epidermis

it is surely related to the demonstrated requirement for epithelial Wnt signalling^[11] and the over-expression of *Wnt7a* in the epidermis generating more hairs,^[10] both results observed in the *Mus* WIHN model. This suggests that *A. cahirinus* uses the same epithelial Wnt pathway. These new hairs rapidly grow down into the newly formed papillary dermis (Figure 1F,G) and form complete new hair follicles including new sebaceous glands (Figure 1H) and erector pili smooth muscles. At the base of the mature hairs, adipocytes begin to differentiate (Figure 1I). Strangely, the new hairs initially appear in the centre of the wound as is the case for the WIHN model,^[10, 11] but in *A. cahirinus* new hairs continue to be induced centrifugally, eventually meeting the old hairs at the original wound edge. At a comparable stage, the CD-1 wound shows a smooth hairless epithelium over a densely collagenous scar (Figure 1J).

3.2 | Response of the dermal fibroblasts

The granulation tissue which fills the hole created in the skin consists of a host of different cell types which invade from the edges and underneath the wound and is very different between these scarring and regenerating species. In CD-1, the tissue is dense from the very beginning and rapidly generates a collagenous matrix in which the cells are randomly orientated. In *A. cahirinus* the granulation tissue is less dense from the outset, stains less intensely for collagen and the cells are orientated with their long axes parallel to the surface of the wound suggesting major differences in matrix production (see below). A denser matrix may have been laid down by more cells and, indeed there are twice as many CD-1 fibroblasts present per unit area for the first 2 weeks compared to *A. cahirinus* and although the proliferation of CD-1 and *A. cahirinus* fibroblasts is initially indistinguishable, *A. cahirinus* cells continue proliferation for a more extended period. Again a denser matrix may contribute to this proliferation difference.

The collagenous nature of the wound bed is widely regarded as being responsible for the scarring that is seen in adults because it has a very different composition from embryonic skin which heals without scarring. In the embryo there is a higher ratio of collagen III to collagen I, higher collagen V, higher levels of hyaluronic acid, decreased levels of many proteoglycans and a low ratio of matrix metalloproteases (MMPs) to their inhibitors (TIMPs).^[22–25] It is interesting to consider how many of these embryonic characteristics are seen in *Acomys* skin regeneration and the answer is many of them but by no means all. For example our RNA-seq analyses have shown that collagen I is not the major contributor to the CD-1 wound since there are higher levels in the *A. cahirinus* day 14 wound. The CD-1 wound shows low level upregulation of collagens IV, VII, VIII, XI, XIV, XXIV, but the highest upregulation is seen with collagen XII (52-fold compared to day 0) and this collagen is strongly present by immunofluorescence precisely in the CD-1 wound bed.^[19] This does not mean that total collagen levels are the highest in the CD-1 wound because we see a 150-fold upregulation of collagen XXIV in *A. cahirinus* relative to day 0. Another striking feature of the *A. cahirinus* wound is a 40-fold upregulation of the collagen triple helix repeat

containing 1 (*Cthrc1*) gene whose function is to promote migration, reduce collagen deposition and inhibit TGF β signalling. This suggests that not all collagens are scar-inducing, as we might expect collagen XII to be, but there may be an appropriate balance which reflects the scarring vs the regenerating fibroblast.

Cthrc1 may thus contribute to maintaining the fluidity of the wound matrix as the MMPs and TIMPs are known to do. In the CD-1 wound, there are low levels of *Mmp* gene expression such as *Mmp2* and *Mmp9* but high levels of *Timp1* whereas the converse is the case in *A. cahirinus*—*Mmp3*, 9, 12 and 13 are highly upregulated whereas the *Timps* are not. Thus, the embryonic principle of a low ratio of MMPs to TIMPs is certainly seen in *A. cahirinus*. Other ECM genes that are strikingly upregulated in the *A. cahirinus* wound compared to CD-1 are tenascin C and N, fibronectin 1, laminin α 1, fibrillin 2 and aggrecan. These may play a positive role in maintaining a “regenerative” matrix. Although we know very little about how such a matrix may have a positive benefit, the biomechanical properties of matrices play a crucial role in dictating cell behaviours and we have recently considered this aspect of CD-1 vs *A. cahirinus* comparisons.^[21]

3.3 | Response of the adipocytes

In the tissue which regenerates whether it is a collagenous scar (CD-1) or a regenerate (*A. cahirinus*) there are initially no new adipocytes despite their prevalence in the original skin (Figure 1A,B). These cells are left behind at the wound margin and remain there. The dermal tissue in CD-1 scar never generates any new adipocytes, but after 7–8 weeks new adipocytes appear in the *A. cahirinus* regenerated skin. Strikingly, these new adipocytes appear at the base of the new hairs which have regenerated (Figure 1I). The origin of these adipocytes is unknown, but in the large wounds of C57BL/6 mice undergoing WIHN which can produce new hairs the adipocytes also appear at the base of hairs where they are induced from myofibroblasts under the influence of bone morphogenetic protein produced by the hair follicles.^[26] This is another striking similarity between *Acomys* and *M. musculus* regenerative behaviour.

3.4 | Response of the panniculus carnosus

This skeletal muscle layer is located within the hypodermis, between the dermal adipose tissue and the adventitia of the underlying connective tissue.^[27] This tissue, while largely vestigial in humans, is ubiquitous in mammals and allows for twitching movement of the skin as well as thermoregulation^[28] and is well developed in both *M. musculus* and *Acomys* (Figure 1A,B). The panniculus carnosus has a higher level of cell turnover during muscle homeostasis, as well as a higher frequency of incorporation of bone marrow-derived cells, compared to limb skeletal muscles.^[29] While skeletal muscle is one of the few tissues that can regenerate throughout life, it was thought that the presence of the existing extracellular matrix template of the injured muscle was an absolute requirement for regeneration. This has also been demonstrated to be true of panniculus carnosus as well, where after full-thickness excisional wounding, the severed

myofibres of the panniculus carnosus undergo repair by proliferation and differentiation of satellite cells, but no new myofibres regenerate in the wound area.^[29] However, in contrast to previous dogma, *Acomys* appears to be capable of *de novo* regeneration of the panniculus carnosus, in the absence of an existing ECM template. Between 14 and 18 days after full-thickness excisional wounding, *A. cahirinus* reactivates the myogenic programme that occurs during developmental myogenesis, by upregulating the myogenic regulatory factors (MRFs), with expression of *Myf5* and *MyoD1* returning to levels observed in normal skin, or slightly elevated, while *Myog* is upregulated 5- to 10-fold. This increase in *Myog* expression drives an induction of the embryonic myosin *Myh3*, with expression increasing from ~20-fold in day 7–10 wounds to 250-fold in day 18, 450-fold in day 21 and 200-fold in day 28 wounds. In contrast, the MRFs in CD-1 remain downregulated or unchanged in all time points assayed and *Myh3* is only modestly upregulated (from 5- to 30-fold) in day 7, 10 and 21 wounds. Both species exhibit modest upregulation of *Myh3* before the induction of the MRFs and may represent repair of the severed muscle fibres at the wound margins. It is interesting to note the MRFs are not upregulated until sometime around 18 days following injury. It is likely that during this initial period, the ECM is being deposited and remodelled in both species; however, in *A. cahirinus* the ECM is presumably remodelled in a manner that is conducive to muscle regeneration, while in CD-1, the ECM is remodelled into a dense collagenous matrix that is inhibitory to regeneration. This is particularly intriguing given that *Wnt2*, a Wnt thought to be involved in fibrosis,^[30] is upregulated in CD-1 in a pattern very similar to that of the MRFs and *Myh3* in *A. cahirinus*. Thus during the period when muscle regeneration could occur there is instead an upregulation of pro-fibrotic genes that lay down a fibrotic matrix that physically blocks muscle regeneration in CD-1. It is therefore conceivable that, in the absence of a fibrotic response, mammals retain the capacity for regeneration in adulthood that is observed in fetal tissue.

After burn injury, the connective tissue of the panniculus carnosus remains in place (although presumably somewhat physically altered) and so we might expect this layer in CD-1 to regenerate. It does indeed regenerate but by no means well and a considerable amount of fibrosis is seen in contrast to the *A. cahirinus* burn injury where the panniculus carnosus regenerates perfectly.^[20]

3.5 | Immune response

A blunted immune system response is one explanation for the regenerative ability of embryonic skin^[2] and lower vertebrates.^[31] In agreement with these ideas, the immune response to wounding is quite different between *Acomys* and *M. musculus*. The typical mammalian response, as observed in CD-1, initially includes a massive upregulation of pro-inflammatory cytokines and chemokines. This upregulation begins shortly after wounding and peaks between 3 and 5 days, before slowly returning to normal levels of expression.^[18,19] RT-qPCR, microarray and RNA-Seq analyses demonstrate a clear inflammatory response in CD-1, with induction of chemokines and cytokines such as *Cxcl3*, *Csf3*, *Cxcl5* and *Il1 β* , among others. Pathway

analysis of microarray data for day 7 vs normal skin also reveals an enrichment of cytokine and chemokine related pathways.^[18] These data have been confirmed at the protein level by ELISA and Western blot analyses as well. In complete contrast to this, the inflammatory response in *Acomys* appears to be rather blunted in comparison. While some chemokines and cytokines are upregulated in *A. cahirinus* it is to a much lower degree than that observed in CD-1. There is also an upregulation of anti-inflammatory cytokines in *A. cahirinus* wounds, for example, *Il10* and *Cxcl12* which may play a role in dampening the immune response.

The macrophage response to injury is also quite different between *A. cahirinus* and *M. musculus*. While early CD-1 wounds are infiltrated with large numbers of both M1 and M2 macrophages, the early wound bed and regenerating dermis of *A. cahirinus* displays a distinct lack of macrophages. While macrophages are present in similar numbers to CD-1 in the underlying fascia and at the wound margins, the *A. cahirinus* dermis displays very little to no staining of F4/80, Iba-1 and cd206 positive macrophages up until about day 7 or 10 after wounding.^[19] This is similar to the observation by Simkin et al.^[32] that the blastema of regenerating ear holes in *A. cahirinus* is nearly devoid of classically activated macrophages. Given the positive roles of pro-reparative M2 macrophages during wound healing, it is likely that there is an as yet unidentified M2 macrophage subtype within the early regenerating *A. cahirinus* dermis.

An interesting unanswered question is how *Acomys* avoids infection following injury, particularly the loss of large pieces of skin after escaping predation in the wild. Presuming that the weak skin of *Acomys* evolved as a form of autotomy, this would only be of benefit if the animal could avoid succumbing to infection from large surface area open wounds. It is therefore interesting to note that, following injury, *A. cahirinus* displays an increase in expression of a number of genes in the complement pathway. *A. cahirinus* upregulates 13 complement factor or complement component genes 7 days after injury to a greater degree than in CD-1, while CD-1 only upregulates two complement genes compared to *A. cahirinus*. The top upregulated gene between *A. cahirinus* and CD-1 in day 7 wounds is the complement component C6 (~1300 fold increase). Some of the other top upregulated genes between *A. cahirinus* and CD-1 include genes known to be involved in the innate immune response as well, such as pannexin,^[33] the pentraxins *Nptx1* and *Nptx2*,^[34] *fincolin*,^[35] among others. It is also interesting to note that the second most upregulated gene between *A. cahirinus* and CD-1 is preproenkephalin (*Penk*), a gene thought to be vital in pain perception and analgesia. *Penk* is upregulated ~900-fold higher in *A. cahirinus* than in CD-1, and may proffer an advantage after skin loss due to predation.

4 | DISCUSSION

We can see above that there are many molecular differences between the scarring *M. musculus* and the regenerating *Acomys* skin that can be identified. Some of them are produced by the epidermal

cells, dermal fibroblasts and other cell types at the site of damage such as matrix proteins and growth factors. We know, for example that the epidermis of these two species migrates at different rates and their cultured fibroblasts produce different secretomes. Others differences can be attributed to the reaction of their different hematopoietic systems in terms of macrophages and immune cytokines and their response to wounding. It will be of importance in the future to design experiments to determine the relative contribution of each of these components to answer questions about the role of the systemic environment (the hematopoietic system) vs intrinsic genetic differences in dermal fibroblasts in regeneration vs scarring. Elaborating these issues will hopefully give us significant insights into the translational problem of how to prevent scarring and induce the regeneration in human skin.

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CONFLICT OF INTEREST

The authors have declared no conflicting interests.

AUTHOR CONTRIBUTION

MM and JOB performed the majority of the experiments derived from the Maden laboratory and both authors contributed equally to the preparation of the manuscript.

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